GREEN LIFE - HAVE KNOWLEDGE. PAY ATTENTION.

Green Life Startup Package

1. Executive Summary

Product

Green Life is a mobile app that helps users identify the environmental impact of products and materials around them. Users scan a barcode, take a photo, or search by name to view a source-backed impact summary, including carbon footprint, material composition, recyclability, disposal guidance, and confidence level.

Problem

Environmental impact information is difficult to access, fragmented across EPA resources, EPD databases, manufacturer disclosures, and other sustainability datasets. Most consumers cannot easily determine whether a product is environmentally harmful, recyclable, or lower impact than alternatives.

Solution

Green Life converts fragmented environmental data into a fast, mobile-first experience. It identifies products or materials, maps them to known records or estimates, and presents transparent, source-cited environmental metrics.

Why Now

- Consumer demand for sustainability transparency is increasing.
- Public environmental datasets are more available than before, but still hard to use.
- Camera, OCR, and barcode workflows are mature enough for a strong MVP.
- Trustworthy environmental tooling for ordinary consumers remains limited.

Target Market

Initial users:

- environmentally conscious consumers
- homeowners/renters
- shoppers comparing products
- people making disposal or purchase decisions

Future users:

- educators
- sustainability programs
- small business procurement teams

Core Value Proposition

"Scan something and understand its environmental impact in seconds."

MVP

The first release should focus on:

- barcode scan
- text search
- product/category matching
- impact summary
- source attribution
- confidence labeling
- saved history

Business Potential

Possible long-term monetization:

- affiliate recommendations for lower-impact alternatives
- premium consumer features
- B2B sustainability lookup API
- enterprise procurement or reporting tools

2. MVP Feature Spec

Feature 1: Barcode Scan

Purpose:

Allow fast identification of packaged products.

Requirements:

- User can open camera and scan barcode.
- App sends barcode to backend.
- Backend attempts exact product match.
- If found, app returns product profile with environmental data.
- If not found, app offers search fallback.

Success Metric:

- high scan-to-result completion rate on supported products

Feature 2: Text Search

Purpose:

Allow lookup when barcode is missing or scan fails.

Requirements:

- User can search by product name, brand, material, or category.
- Search returns best-match results ranked by relevance.
- User can select a result and open detail view.

Success Metric:

- high rate of successful product/category discovery from search

Feature 3: Photo Capture / Image Input

Purpose:

Support future object identification and OCR-assisted matching.

MVP Scope:

- capture/upload image
- optionally run OCR for visible labels/text
- use image as a fallback input, not primary matching method

Success Metric:

- assists search resolution when barcode is unavailable

Feature 4: Impact Summary Screen

Purpose:

Present environmental information clearly and quickly.

Requirements:

- Show product/item name
- Show match type: exact / probable / estimate
- Show key metrics:
- carbon footprint
- material composition
- recyclability
- recycled content
- disposal guidance
- hazard flags where available
- Show source attribution for each metric
- Show freshness/date where available

Success Metric:

- users understand the result without needing to open technical details

Feature 5: Confidence Labeling

Purpose:

Prevent overstating precision.

Requirements:

- Every result must display one of:
- Exact Product Match
- Probable Product Match
- Category Estimate
- Material Estimate
- Confidence label must appear near top of result screen.
- Metrics derived from estimates should be visibly marked.

Success Metric:

- low user confusion and low trust erosion from uncertain data

Feature 6: Saved Items / History

Purpose:

Improve retention and repeat use.

Requirements:

- User can save an item
- User can view prior scans/searches
- User can remove saved entries

Success Metric:

- repeat engagement and retained value after first use

Feature 7: Admin Review Tool

Purpose:

Support internal quality control.

Requirements:

- Admin can inspect product-source mappings
- Admin can review questionable matches
- Admin can correct category/product associations
- Admin can view data provenance and confidence metadata

Success Metric:

- reduced error rate in user-facing results

3. Technical Architecture

Architecture Overview

Use a mobile client backed by a normalization and matching platform.

Recommended stack:

- Mobile: React Native with Expo
- Backend API: FastAPI or NestJS
- Database: PostgreSQL
- Search: PostgreSQL full-text initially
- Storage: S3-compatible object storage
- Auth: Clerk, Firebase Auth, or Auth0
- Admin UI: simple React web panel

Core Components

Mobile App

- onboarding/auth
- barcode scanner
- camera/photo upload
- search UI
- result screen
- saved history

API Layer

- authentication
- product lookup endpoints
- search endpoints
- saved item endpoints
- source/metric retrieval

Matching Service

- barcode exact match
- search ranking
- OCR-assisted product matching
- fallback category/material estimation

Environmental Scoring Service

- retrieves normalized impact metrics
- resolves conflicts between sources
- attaches confidence score
- formats results for API clients

Data Ingestion / ETL

- imports EPA and other datasets
- parses EPD and structured disclosures
- maps incoming records to canonical schema
- stores provenance and freshness data

Admin Tooling

- review records
- inspect source lineage
- override mappings
- QA uncertain results

Canonical Data Model

Core tables/entities:

- products
- brands
- categories
- materials
- product_identifiers
- impact_metrics
- sources
- source_records
- product_matches
- saved_items
- users

Lookup Flow

1. User scans barcode or searches text.

2. API sends request to matching service.
3. Matching service resolves product or fallback category/material.
4. Scoring service retrieves environmental metrics.
5. API returns normalized response with:

- item identity
- metrics
- source attribution
- confidence label
- timestamps/freshness

Data Principles

- preserve source provenance
- never discard methodology metadata if available
- separate exact data from estimates
- track freshness and dataset version
- support partial records without breaking UX

External Data Integration Strategy

Integrate in layers:

6. Product metadata/barcode source
7. EPA/public environmental datasets
8. EPD providers
9. Manufacturer disclosures
10. Category-level carbon/material factor datasets

Primary Technical Risk

The main challenge is not UI. It is data normalization, confidence scoring, and source quality management.

4. User Stories and Acceptance Criteria

Epic: Product Identification

Story 1

As a user, I want to scan a barcode so I can identify a product quickly.

Acceptance Criteria:

- User can open barcode scanner from home screen.
- Valid barcode returns a result if product exists in system.
- If no exact match is found, user is shown a fallback path.
- Result loads within acceptable performance target under normal conditions.

Story 2

As a user, I want to search by name or brand so I can find products manually.

Acceptance Criteria:

- Search input accepts free text.
- Search results are ranked by relevance.
- User can open a result detail page from search results.
- Empty or failed search states provide recovery guidance.

Epic: Environmental Impact Display

Story 3

As a user, I want to see a product's carbon footprint and related metrics.

Acceptance Criteria:

- Result screen shows available environmental metrics.
- Each metric includes unit and source.
- Missing metrics do not break the page.
- Metrics are clearly labeled as exact or estimated when relevant.

Story 4

As a user, I want to know how reliable the result is.

Acceptance Criteria:

- Each result displays match/confidence type.
- Confidence label appears before detailed metrics.
- Estimated results are visually differentiated from exact matches.

Epic: Transparency and Trust

Story 5

As a user, I want to know where the data came from.

Acceptance Criteria:

- Each key metric includes source attribution.
- Source metadata is accessible in the detail view.
- Dataset date/version is shown where available.

Story 6

As a user, I want the app to avoid false precision.

Acceptance Criteria:

- The system never presents category estimates as exact product values.
- Fallback estimates are labeled in API and UI.
- Confidence logic is enforced consistently across supported result types.

Epic: Retention

Story 7

As a user, I want to save items for later.

Acceptance Criteria:

- Signed-in users can save a result.
- Saved items appear in a history/list view.
- Users can remove saved items.

Epic: Admin and Data Quality

Story 8

As an admin, I want to review product-to-source mappings.

Acceptance Criteria:

- Admin can search products and inspect matched sources.
- Admin can see confidence metadata and source lineage.
- Admin can flag or correct problematic mappings.

Story 9

As an admin, I want to inspect ingest quality.

Acceptance Criteria:

- Admin can view imported source records.
- Admin can see ingestion date and source name.
- Failed or incomplete mappings are visible for review.

5. Suggested MVP Backlog

Sprint 1

- finalize supported categories
- define schema
- set up mobile app shell
- set up backend, auth, database

Sprint 2

- barcode scan flow
- text search flow
- product/result API
- initial admin panel scaffold

Sprint 3

- ingest first product metadata source
- ingest first environmental data source

- implement canonical metric model
- build result detail UI

Sprint 4

- confidence labeling
- saved items/history
- QA and admin correction flow
- MVP analytics and error tracking

Sprint 5

- hardening
- usability testing
- data coverage review
- launch prep

6. Recommended MVP Narrowing

If you want the highest chance of shipping, start with one of these:

- packaged household products
- food and beverage packaged goods
- building materials and furniture

Do not start with "everything around you." That creates a computer vision and data coverage problem too early.

7. Immediate Next Deliverables

Possible next artifacts:

- investor pitch deck outline
- app screen map and wireframe spec
- database schema
- API endpoint spec
- a phased build plan with team roles and timeline

The most useful next step after this package was the screen map and API spec, because that turns the concept package into a buildable product.